

GymOS

Gym Management & Member Operations Platform

Complete Product Requirements Document

Version 2.0 • Commercial Product Specification

A production-oriented blueprint covering product modules, subscription tiers, workflows, security, WhatsApp automation, architecture, costs, and roadmap.

Master document for product design, engineering, testing and commercial launch.

1. Executive Summary

GymOS is an all-in-one gym management platform designed to digitize and automate the daily operations of gyms, fitness centers and personal training studios. It manages the complete lifecycle of a member: Lead → Registration → Membership → Payment → Check-in → Workout → Attendance → Communication → Renewal → Retention.

The product is designed as a commercial SaaS, with Basic, Pro, Business and Enterprise tiers. The architecture should support strict tenant isolation, reliable backups, role-based access and future multi-branch expansion.

2. Product Vision

Build an affordable, reliable and intelligent operating system for gyms.

GymOS should help owners answer who their members are, who entered today, whose membership is expiring, who has paid, how much revenue was generated, which leads have not converted, which members are inactive, and what operational actions need attention.

3. Problem Statement

- Paper registers and spreadsheets create inconsistent records.
- Payment and membership data may be disconnected.
- Renewal opportunities can be missed.
- Attendance is difficult to analyze manually.
- Owners lack centralized business visibility.
- Member communication is repetitive.

Solution

One platform for Members + Attendance + Membership + Payments + Trainers + Workouts + Leads + Communication + Analytics.

4. Target Customers

- Small gyms: approximately 50–300 members.
- Medium gyms: approximately 300–1,000 members.
- Fitness centers with multiple trainers and services.
- Gym chains with multiple branches.

5. User Roles

Role	Main Capabilities
Owner	Full gym settings, members, staff, trainers, payments, reports, subscriptions, integrations, branches
Manager	Members, attendance, payments, memberships, leads, trainers, reports
Receptionist	Registration, search, check-in/out, payments, renewals, receipts, leads
Trainer	Assigned members, workouts, sessions, measurements, appointments
Member	Membership, attendance, workout, trainer, payments, progress, appointments

6. Subscription Model

GymOS will use a subscription model. Prices below are initial product-design targets and should be validated with real gym owners before launch.

Plan	Suggested Price	Target	Key Features
Basic	■499–■699/mo	Small gyms	Up to 100 active members; members, plans, payments, QR attendance, entry/exit, expiry, invoices, b
Pro	■999–■1,499/mo	Growing gyms	Everything Basic + up to 500 members, automated WhatsApp, trainers, workouts, progress, CRM, tri
Business	■1,999–■2,999/mo	Professional centers	Everything Pro + AI assistant, advanced analytics, marketing automation, custom branding, API, adv
Enterprise	Custom	Gym chains	Multi-branch, centralized analytics, custom integrations, dedicated infrastructure options, enterprise s

7. Subscription Feature Matrix

Feature	Basic	Pro	Business	Enterprise
Members	100	500	Custom	Custom
Dashboard	✓	✓	✓	✓
Membership	✓	✓	✓	✓
Payments	✓	✓	✓	✓
QR Check-in	✓	✓	✓	✓
Attendance	✓	✓	✓	✓
Invoice	✓	✓	✓	✓
WhatsApp Click-to-Chat	✓	✓	✓	✓
Automated WhatsApp	—	✓	✓	✓
Trainers	—	✓	✓	✓
Workout Plans	—	✓	✓	✓
Body Progress	—	✓	✓	✓
CRM / Leads	—	✓	✓	✓
Appointments	—	✓	✓	✓
Inventory	—	✓	✓	✓
Advanced Reports	—	✓	✓	✓
AI Assistant	—	—	✓	✓
Marketing Automation	—	—	✓	✓
Multi-Branch	—	—	Optional	✓
API	—	—	✓	✓
Custom Branding	—	—	✓	✓

8. Dashboard

- Owner KPIs: total, active, expired and expiring members; today's check-ins; currently inside; today's and monthly revenue; new members; renewals; pending payments; leads; inactive members.
- Quick actions: Add Member, Check-in, Check-out, Payment, Renewal, Add Lead, Add Trainer.
- Recent activity: registrations, payments, renewals, check-ins, check-outs and membership changes.
- Role-based dashboard content.

9. Member Management

- Auto-generated unique Member ID such as GYM-000001.
- Profile: name, photo, mobile, email, DOB, gender, address, emergency contact, registration date, source and notes.
- Tabs: Overview, Membership, Attendance, Payments, Workout, Progress, Trainer, Appointments, Messages and Activity History.

- Search, filtering, editing, deactivation and full historical view.

10. Membership Management

- Custom plan name, duration, price, discount, tax, benefits, freeze days, trial availability and status.
- Support monthly, quarterly, half-yearly, yearly, personal training and custom plans.
- Statuses: ACTIVE, EXPIRING SOON, EXPIRED, FROZEN, CANCELLED, TRIAL.
- Renewals must create a new historical membership period rather than overwrite the old one.

11. Attendance & Entry/Exit

V1: manual search, Member ID and QR code. Future: RFID, biometric and face recognition, subject to privacy, consent, legal and accuracy requirements.

Scan QR → Find Member → Verify Membership → Active? → Record Entry → Success → Optional Notification

Attendance stores member, date, entry time, exit time, check-in method and device/source.

12. Payment System

- Payment ID, member, amount, plan, discount, tax, payment method, transaction ID, date, receiver and notes.
- Methods: Cash, UPI, Card, Bank Transfer and Other.
- Successful payment updates the relevant membership and creates a payment history record.
- Payment records should be immutable in normal operation; corrections should use authorized adjustments with audit logs.

13. Invoice & Receipt

Payment → Invoice → Receipt → Optional Notification.

- Gym name and address.
- Invoice number.
- Member and plan.
- Amount, discount, tax and final amount.
- Payment method and transaction ID.
- Print and PDF download.

14. Renewal & Expiry Automation

Default reminder rules: 30 days, 7 days, 3 days, 1 day and expiry day. Gym owners can customize rules.

Membership Expiring → Dashboard Alert → Optional WhatsApp Reminder → Renewal → Payment → New Membership Period → Receipt → Confirmation

15. WhatsApp Strategy

Click-to-WhatsApp

Available in Basic. GymOS opens WhatsApp with a pre-filled message; the staff member presses Send. This avoids relying on an automated API for basic communication.

Automated WhatsApp

Available in Pro and above through an authorized WhatsApp Business integration. Use for registration, payment, renewal, expiry, trial expiry, inactivity, birthday and important notifications.

Messaging usage should be tracked separately. Variable third-party messaging costs should not be silently absorbed into unlimited base subscriptions.

16. Trainer Management

- Trainer ID, name, photo, mobile, email, specialization, joining date, status and compensation fields.
- Assign members to trainers.
- Manage sessions, appointments and workout plans.
- Trainer dashboard for assigned members and upcoming/completed sessions.

17. Workout Management

Workout plans contain day, muscle group, exercise, sets, repetitions, weight, rest and notes. Trainers can assign plans to members and update progress.

18. Body Progress

- Weight, height, BMI, body-fat percentage, chest, waist, arms, thighs and custom metrics.
- Historical measurements.
- Progress charts over time.

19. Lead / CRM

New Lead → Contacted → Visited → Trial → Converted → Member

- Lead ID, name, phone, source, interested plan, follow-up date, status, assigned staff and notes.
- Track lead-to-member conversion.
- Support follow-up reminders.

20. Trial Membership

- 1-day, 3-day, 7-day and custom trials.
- Automatic trial expiry.
- Optional reminder before trial ends.
- Conversion of trial into membership while preserving history.

21. Inactive Member Detection

GymOS should identify members who have not visited for configurable periods such as 3, 7, 14 and 30 days. Owners can review and optionally trigger re-engagement communication.

22. Appointment Management

- Personal training and trainer sessions.
- Calendar view.
- Conflict prevention / no double booking.
- Member and trainer association.
- Session status and history.

23. Inventory

- Products, SKU, category, quantity, purchase price, selling price, supplier and low-stock threshold.

- Stock-in and stock-out transactions.
- Low-stock alerts.
- Inventory history.

24. Reports & Analytics

- Member reports: new, active, expired, expiring, renewed, cancelled and inactive.
- Financial: daily, weekly, monthly, custom revenue, payment methods and pending payments.
- Attendance: daily, weekly, monthly, member-wise and peak hours.
- Trainer and appointment reports.
- Inventory reports.
- Export to PDF and CSV/Excel.

25. AI Assistant — Business+

The AI assistant can answer natural-language questions using authorized gym data through a controlled retrieval layer.

Owner: "How many memberships expire this week?"

AI: "27 active memberships are scheduled to expire in the selected period."

- Revenue queries.
- Expiry queries.
- Inactive-member queries.
- Attendance queries.
- Plan distribution queries.
- Trainer and operational queries.

The AI must not invent database values and should state when required information is unavailable.

26. Member Portal

- Dashboard.
- Membership.
- Attendance.
- Workout.
- Trainer.
- Payments.
- Progress.
- Appointments.
- Notifications.

27. Multi-Branch

Enterprise architecture should support Organization → Branch → Members / Staff / Trainers / Payments / Attendance.

- Centralized owner dashboard.
- Branch-level revenue and attendance.
- Branch-level staff and permissions.
- Cross-branch reporting.

- Future cross-branch member rules.

28. GymOS Subscription Billing

GymOS itself must manage customer subscriptions.

Free Trial → Plan Selection → Payment → Active Subscription → Renewal → Upgrade/Downgrade → Cancellation

- Subscription ID.
- Plan.
- Start and renewal date.
- Status.
- Payment status.
- Customer/organization.
- Usage limits.

29. Free Trial

A 7–14 day trial is recommended for validation. Selected Pro capabilities can be exposed during the trial so owners can understand the product before subscribing.

30. Usage Limits

- Limits should be configurable by plan.
- Example: Basic = 100 active members.
- When a limit is reached, show an upgrade prompt rather than deleting or blocking access to historical data.
- Historical data remains preserved after downgrade or cancellation subject to the product's retention policy.

31. Security

- Secure password hashing.
- HTTPS for networked deployments.
- Role-based access control.
- Server-side authorization on protected APIs.
- Input validation.
- Rate limiting where appropriate.
- Secure session/token handling.
- Audit logs.
- Backups and recovery.
- Strict tenant isolation.

32. Tenant Isolation

GymOS must be multi-tenant. Each organization must only access its own data. Tenant boundaries must be enforced server-side, not only through frontend filters.

Gym A → only Gym A Members / Payments / Attendance / Reports
Gym B → only Gym B Members / Payments / Attendance / Reports

33. Audit Logs

Important actions should record user, action, resource, timestamp and relevant metadata.

10:31 AM – Receptionist added Member GYM-00241

10:45 AM – Manager changed membership

11:02 AM – Owner recorded ■4,500 payment

34. Backup & Restore

- Automatic scheduled backup.
- Manual backup.
- Restore capability for authorized administrators.
- Backup history.
- Backup status.
- Recovery testing as part of operations.

35. Data Export

- CSV.
- Excel.
- PDF.
- Member data.
- Payment data.
- Attendance.
- Reports.

36. Notification Center

- Expired memberships.
- Expiring memberships.
- Pending payments.
- New leads.
- Low-stock items.
- System/backup alerts.

37. System Settings

- Gym name, logo, address and contact.
- Currency and regional settings.
- Membership settings.
- Attendance settings.
- Notification settings.
- WhatsApp settings.
- Receipt settings.
- Staff permissions.
- Backup settings.

38. Branding

Business and Enterprise tiers may support gym logo, brand name, receipt branding, member-portal branding and custom communication templates. White-labeling can be considered as an enterprise feature.

39. Recommended Technical Architecture

```
Frontend: React / Next.js
Backend: FastAPI
Database: PostgreSQL
Authentication: Secure session/JWT architecture
Storage: Object storage for photos, documents and backups
Messaging: Authorized WhatsApp Business integration
AI: LLM + controlled database retrieval layer
Deployment: Shared multi-tenant cloud infrastructure
Optional future: local/offline-capable deployment
```

40. Core Database Entities

```
users
roles
permissions
organizations
branches
members
membership_plans
memberships
attendance
payments
invoices
trainers
trainer_assignments
workout_plans
workout_exercises
body_measurements
leads
appointments
inventory
inventory_transactions
notifications
message_templates
message_logs
audit_logs
subscriptions
subscription_plans
subscription_payments
backups
settings
```

41. Critical Business Rules

- Expired membership is automatically identified from its validity dates.
- Check-in is allowed when membership is ACTIVE, subject to authorized gym policy overrides.
- Exit updates the open attendance record.
- Successful payment updates the relevant membership, creates the invoice/receipt and audit record, and triggers configured notification.
- Appointments cannot be double-booked.
- Historical memberships and payments are preserved.
- Sensitive administrative actions are auditable.

42. New Member Workflow

```
Add Member
↓
Select Plan
```

```
↓
Payment
↓
Membership Created
↓
Member ID
↓
QR Code
↓
Receipt
↓
Welcome Notification
↓
ACTIVE
```

43. Check-in Workflow

```
QR / Member ID
↓
Find Member
↓
Verify Membership
↓
ACTIVE?
↓ ↓
YES NO
↓ ↓
Entry Show
Recorded Expired
```

44. Renewal Workflow

```
Membership Expiring
↓
Reminder
↓
Member Renews
↓
Payment
↓
New Membership Period
↓
Receipt
↓
Confirmation
```

45. Development Roadmap

Phase 1 — MVP

- Authentication and roles.
- Dashboard.
- Member management.
- Membership plans.
- Payments.
- QR attendance.
- Entry/exit.
- Expiry tracking.
- Receipts.
- Basic reports.

- Audit logs.
- Backup/restore.
- Multi-tenant foundation.
- Subscription foundation.

Phase 2 — Pro

- Automated WhatsApp.
- Trainers.
- Workouts.
- Body measurements.
- CRM.
- Trials.
- Appointments.
- Inactive-member detection.
- Inventory.
- Advanced reports.
- Multiple staff accounts.

Phase 3 — Business

- AI assistant.
- Advanced analytics.
- Marketing automation.
- Member portal.
- Custom branding.
- API integrations.
- Advanced permissions.

Phase 4 — Enterprise

- Multi-branch.
- Centralized management.
- Branch analytics.
- Enterprise integrations.
- White-label.
- Dedicated infrastructure options.

46. Cost-Control Strategy

- Use open-source technologies where practical.
- Develop and test locally before paying for production services.
- Use shared multi-tenant infrastructure rather than one server per gym.
- Start with one appropriately sized production environment and scale only when usage requires it.
- Keep AI, automated WhatsApp and other variable-cost services behind paid tiers or usage-based billing.
- Do not build expensive hardware integrations in V1.

47. WhatsApp Cost Strategy

- Use click-to-chat for Basic where staff can press Send.
- Reserve automated messaging for paid tiers.
- Do not send unnecessary messages for every gym entry by default.
- Prioritize registration, payment, renewal, expiry, trial and important alerts.
- Track message volume and status.
- Treat third-party messaging charges transparently and separately where applicable.

48. Example Unit Economics

If GymOS eventually serves 100 gyms at an average subscription of █999/month, gross subscription revenue would be approximately █99,900/month before taxes, payment processing, infrastructure, support, messaging, AI and other business expenses.

The multi-tenant architecture is intended to reduce average infrastructure cost per gym as customer count grows, while maintaining strict tenant isolation.

49. UX Requirements

- Simple and fast interface.
- Laptop-friendly for reception desks.
- Responsive web experience.
- Minimal training required.
- Common actions should take only a few clicks.
- Prioritize Search → Check-in → Payment → Member.

50. Production Acceptance Criteria

- Member creation/editing works reliably.
- Membership dates and statuses calculate correctly.
- Payments are persisted correctly.
- Receipts generate correctly.
- QR check-in works.
- Entry/exit records work.
- Expiry tracking is accurate.
- Search and filters work.
- Role permissions are enforced server-side.
- Tenant isolation is verified.
- Backups can be created and restored.
- Audit logs work.
- Critical errors are handled and logged.
- Historical records are preserved.

51. Future Hardware

- RFID card and reader integration.

- Biometric entry.
- Camera-based recognition, subject to privacy, consent, legal and accuracy requirements.

52. Product Success Metrics

- Active gyms.
- Active staff users.
- Members managed.
- Daily check-ins.
- Payment transactions.
- Membership renewals.
- Lead conversions.
- Automated messages.
- System uptime.
- Backup success rate.
- Customer retention.
- Plan upgrades and cancellations.

53. Final Product Principle

Every gym action should create useful data, and that data should help the gym owner take the next useful action.

```
Member stops visiting
↓
GymOS detects inactivity
↓
Owner gets alert
↓
Re-engagement message
↓
Member returns
↓
Attendance recorded
↓
Renewal opportunity
```

54. Final MVP Definition

```
GYMOS V1

✓ Login
✓ Owner / Manager / Receptionist roles
✓ Dashboard
✓ Member Management
✓ Membership Plans
✓ Payments
✓ QR Check-in
✓ Entry / Exit
✓ Attendance History
✓ Expiry Tracking
✓ Renewal
✓ Invoice / Receipt
✓ Basic WhatsApp Click-to-Chat
✓ Basic Reports
✓ Search / Filters
✓ Audit Logs
✓ Backup / Restore
✓ Multi-tenant foundation
✓ Subscription foundation
```

The first priority is reliability: a real gym must be able to use V1 every day without losing, corrupting or mixing customer data. Higher-tier features should be modular so Basic → Pro → Business upgrades can be enabled without rewriting the core system.